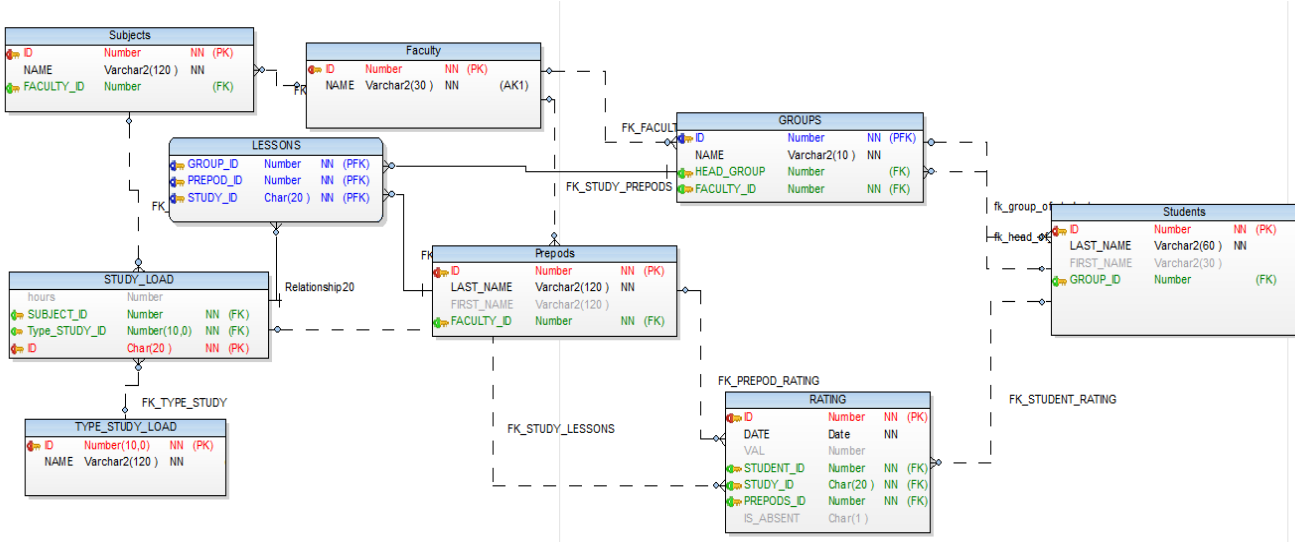
Лабораторная работа №3
Проектирование схемы базы данных в SQL

Цель работы: Получить практические навыки построения баз данных с помощью диалекта DDL языка SQL.

Задача 1. Разобраться со схемой базы данных информационной системы организации учебного процесса, которая представлена на следующем рисунке.



Общие сведения. Схема отображает структуру базы данных информационной системы для организации учебного процесса.

Таблица «Faculty» Справочник факультетов.

Таблица «GROUPS» Справочник групп.

Таблица «LESSONS» справочник занятий, объединяющий преподавателя, предмет и вид учебной нагрузки

Таблица «TYPE_STUDY_LOAD»справочник типов учебной нагрузки (лекции, практические занятия, семинары и др.). и т.д. Сгенеррированный отчет см. Приложение

Задача. Разработка схемы базы данных .

Таблицы представляют собой главный инструмент моделирования данных в *БД*, как построенных на основе *SQL* вообще, так и в*Oracle* в частности.

Предложение CREATE TABLE

Создание таблиц осуществляется предложением CREATE TABLE категории *DDL*.

```
CREATE TABLE таблица
(
    поле1 тип ограничение,
    поле2 тип ограничение,
    поле3 тип ограничение
);
```

Пример:

```
CREATE TABLE proj
(
   projno NUMBER   ( 4 )
 , pname  VARCHAR2 ( 14 )
 , bdate  DATE
 , budget NUMBER   ( 10, 2 )
);
```

Помимо указанных в этом примере предложение CREATE TABLE может включать много других конструкций, большая часть которых связана с организацией хранения, а не с логикой данных. Полностью они приводятся в документации Oracle по SQL.

Типы данных в столбцах

Тип данных	Описание
Строковые типы	
VARCHAR2(длинна)	Строка переменной длинны (до 2 000)
CHAR(длинна)	Строка фиксированной длинны (до 255)
Числовые типы	
NUMBER(p,s)	число переменной длинны
Типы дата-время	
DATE	Дата время с точностью до секунды
TIMESTAMP	Дата время с точностью до долей секунды в Unix-формате

Создание связей

CONSTRAINT имя_оганичения REFERENCES Таблица_на_которую ссылаемся(Столбец_на который_ссылаемся)

В результате выполнения работы вы должны иметь набор SQL-запросов позволяющий создать базу данных согласно предложенной схеме.

Entity 'Faculty'

Caption	Faculty

Primary Key 'PK_FACULTY'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number	YES	NO	NO	NO	Первичный ключ
	NAME		Varchar2(30)	YES	YES	NO	NO	Название факультета

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_STUDY_PREPODS	Non-identifying	Faculty	Prepods	1:N
FK_FACULTY_GROUP	Non-identifying	Faculty	GROUPS	1:N
FK_FACULTY_SUBJECTS	Non-identifying	Faculty	Subjects	1:N

Keys

Key Type	Constraint Name	Attributes
PK	PK_FACULTY	ID
AK	NAME	NAME

Comments

Справочник факультетов

Attribute 'ID'

Caption	ID
Entity Name	Faculty
Primary Key	YES
Foreign Key	NO
Data Type	Number
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'NAME'

Caption	NAME
Entity Name	Faculty
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(30)
Not Null	YES
Unique	YES

Comments

Название факультета

Entity 'GROUPS'

Caption	GROUPS

Primary Key 'pk_group'

Attribute Name	type	Parent Entity
ID	PFK	Students

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PFK	ID		Number	YES	NO	NO	NO	Первичный ключ
	NAME		Varchar2(10)	YES	NO	NO	NO	Номер группы
FK	HEAD_GROUP		Number	NO	NO	NO	NO	Староста группы
FK	FACULTY_ID		Number	YES	NO	NO	NO	Факультет, к которому принадлежит группа

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
fk_head_of_group	Non-identifying	Students	GROUPS	1:N
fk_group_of students	Non-identifying	GROUPS	Students	1:N
fk_head_of_group	Non-identifying	Students	GROUPS	1:N
FK_FACULTY_GROUP	Non-identifying	Faculty	GROUPS	1:N
FK_GROUP_LESSONS	Identifying	GROUPS	LESSONS	1:N

Comments

Информация о группах

Attribute 'ID'

Caption	ID		
Entity Name	GROUPS		
Primary Key	YES		
Foreign Key	YES	Parent Entity	Students
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Первичный ключ

Attribute 'NAME'

Caption	NAME
Entity Name	GROUPS
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(10)
Not Null	YES
Unique	NO

Comments

Номер группы

Attribute 'HEAD_GROUP'

Caption	HEAD_GROUP		
Entity Name	GROUPS		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Students
Data Type	Number		
Not Null	NO		
Unique	NO		

Comments

Староста группы

Attribute 'FACULTY_ID'

Caption	FACULTY_ID		
Entity Name	GROUPS		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Faculty
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Факультет, к которому принадлежит группа

Entity 'LESSONS'

Caption	LESSONS

Primary Key 'Key3'

Attribute Name	type	Parent Entity
GROUP_ID	PFK	GROUPS
PREPOD_ID	PFK	Prepods
STUDY_ID	PFK	STUDY_LOAD

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PFK	GROUP_ID		Number	YES	NO	NO	NO	Первичный ключ
PFK	PREPOD_ID		Number	YES	NO	NO	NO	Преподаватель
PFK	STUDY_ID		Char(20)	YES	NO	NO	NO	Учебная нагрузка

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_GROUP_LESSONS	Identifying	GROUPS	LESSONS	1:N
FK_PREPODS_LESSONS	Identifying	Prepods	LESSONS	1:N
Relationship20	Identifying	STUDY_LOAD	LESSONS	1:N

Comments

Нагрузка преподавателя

Attribute 'GROUP_ID'

Caption	GROUP_ID		
Entity Name	LESSONS		
Primary Key	YES		
Foreign Key	YES	Parent Entity	GROUPS
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Первичный ключ

Attribute 'PREPOD_ID'

Caption	PREPOD_ID		
Entity Name	LESSONS		
Primary Key	YES		
Foreign Key	YES	**Parent Entity**	Prepods
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Преподаватель

Attribute 'STUDY_ID'

Caption	STUDY_ID		
Entity Name	LESSONS		
Primary Key	YES		
Foreign Key	YES	**Parent Entity**	STUDY_LOAD
Data Type	Char(20)		
Not Null	YES		
Unique	NO		

Comments

Учебная нагрузка

Entity 'Prepods'

Caption	Prepods

Primary Key 'PK_PREPODS'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number	YES	NO	NO	NO	Первичный ключ
	LAST_NAME		Varchar2(120)	YES	NO	NO	NO	Фамилия преподавателя
	FIRST_NAME		Varchar2(120)	NO	NO	NO	NO	Имя преподавателя
FK	FACULTY_ID		Number	YES	NO	NO	NO	Факультет, на котором работает преподаватель

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_STUDY_PREPODS	Non-identifying	Faculty	Prepods	1:N
FK_PREPODS_LESSONS	Identifying	Prepods	LESSONS	1:N
FK_PREPOD_RATING	Non-identifying	Prepods	RATING	1:N

Comments

Справочник преподавателей

Attribute 'ID'

Caption	ID
Entity Name	Prepods
Primary Key	YES
Foreign Key	NO
Data Type	Number
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'LAST_NAME'

Caption	LAST_NAME
Entity Name	Prepods
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(120)
Not Null	YES
Unique	NO

Comments

Фамилия преподавателя

Attribute 'FIRST_NAME'

Caption	FIRST_NAME
Entity Name	Prepods
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(120)
Not Null	NO
Unique	NO

Comments

Имя преподавателя

Attribute 'FACULTY_ID'

Caption	FACULTY_ID		
Entity Name	Prepods		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Faculty
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Факультет, на котором работает преподаватель

Entity 'RATING'

Caption	RATING

Primary Key 'PK_RATING'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number	YES	NO	NO	NO	Первичный ключ
	DATE		Date	YES	NO	NO	NO	Дата получения отметки
	VAL		Number	NO	NO	NO	NO	Полученная отметка
FK	STUDENT_ID		Number	YES	NO	NO	NO	Студент
FK	STUDY_ID		Char(20)	YES	NO	NO	NO	Учебный предмет
FK	PREPODS_ID		Number	YES	NO	NO	NO	Преподаватель
	IS_ABSENT		Char(1)	NO	NO	NO	YES	Отсутсвовал ли студент на занятии

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_STUDENT_RATING	Non-identifying	Students	RATING	1:N
FK_STUDY_LESSONS	Non-identifying	STUDY_LOAD	RATING	1:N
FK_PREPOD_RATING	Non-identifying	Prepods	RATING	1:N

Check Constraints

Check Constraint Name	chkValue
Check	val>3 and val<=10

Comments

Отметки полученные студентами

Attribute 'ID'

Caption	ID
Entity Name	RATING
Primary Key	YES
Foreign Key	NO
Data Type	Number
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'DATE'

Caption	DATE
Entity Name	RATING
Primary Key	NO
Foreign Key	NO
Data Type	Date
Not Null	YES
Unique	NO

Comments

Дата получения отметки

Attribute 'VAL'

Caption	VAL
Entity Name	RATING
Primary Key	NO
Foreign Key	NO
Data Type	Number
Not Null	NO
Unique	NO

Comments

Полученная отметка

Attribute 'STUDENT_ID'

Caption	STUDENT_ID		
Entity Name	RATING		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Students
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Студент

Attribute 'STUDY_ID'

Caption	STUDY_ID		
Entity Name	RATING		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	STUDY_LOAD
Data Type	Char(20)		
Not Null	YES		
Unique	NO		

Comments

Учебный предмет

Attribute 'PREPODS_ID'

Caption	PREPODS_ID		
Entity Name	RATING		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Prepods
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Преподаватель

Attribute 'IS_ABSENT'

Caption	IS_ABSENT
Entity Name	RATING
Primary Key	NO
Foreign Key	NO
Data Type	Char(1)
Not Null	NO
Unique	NO
Default	'N'

Comments

Отсутсвовал ли студент на занятии

Entity 'Students'

Caption	Students

Primary Key 'pk_students'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number	YES	NO	NO	NO	Первичный ключ
	LAST_NAME		Varchar2(60)	YES	NO	NO	NO	Фамилия студента
	FIRST_NAME		Varchar2(30)	NO	NO	NO	NO	Имя студента
FK	GROUP_ID		Number	NO	NO	NO	NO	Группа, в которой обучается студент

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
fk_head_of_group	Non-identifying	Students	GROUPS	1:N
fk_group_of students	Non-identifying	GROUPS	Students	1:N
fk_head_of_group	Non-identifying	Students	GROUPS	1:N
FK_STUDENT_RATING	Non-identifying	Students	RATING	1:N

Comments

Справочник студентов

Attribute 'ID'

Caption	ID
Entity Name	Students
Primary Key	YES
Foreign Key	NO
Data Type	Number
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'LAST_NAME'

Caption	LAST_NAME
Entity Name	Students
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(60)
Not Null	YES
Unique	NO

Comments

Фамилия студента

Attribute 'FIRST_NAME'

Caption	FIRST_NAME
Entity Name	Students
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(30)
Not Null	NO
Unique	NO

Comments

Имя студента

Attribute 'GROUP_ID'

Caption	GROUP_ID		
Entity Name	Students		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	GROUPS
Data Type	Number		
Not Null	NO		
Unique	NO		

Comments

Группа, в которой обучается студент

Entity 'STUDY_LOAD'

Caption	STUDY_LOAD

Primary Key 'PK_STUDY_LOAD'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
	hours		Number	NO	NO	NO	NO	Количество планируемых часов на дисциплину
FK	SUBJECT_ID		Number	YES	NO	NO	NO	Предмет
FK	Type_STUDY_ID		Number(10,0)	YES	NO	NO	NO	Тип нагрузки
PK	ID		Char(20)	YES	NO	NO	NO	Первичный ключ

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_SUBJECT_STUDY	Non-identifying	Subjects	STUDY_LOAD	1:N
FK_TYPE_STUDY	Non-identifying	TYPE_STUDY_LOAD	STUDY_LOAD	1:N
FK_STUDY_LESSONS	Non-identifying	STUDY_LOAD	RATING	1:N
Relationship20	Identifying	STUDY_LOAD	LESSONS	1:N

Check Constraints

Check Constraint Name	chkStudyLoad
Check	hours>=0

Comments

Учебная нагрузка

Attribute 'hours'

Caption	hours
Entity Name	STUDY_LOAD
Primary Key	NO
Foreign Key	NO
Data Type	Number
Not Null	NO
Unique	NO

Comments

Количество планируемых часов на дисциплину

Attribute 'SUBJECT_ID'

Caption	SUBJECT_ID		
Entity Name	STUDY_LOAD		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Subjects
Data Type	Number		
Not Null	YES		
Unique	NO		

Comments

Предмет

Attribute 'Type_STUDY_ID'

Caption	Type_STUDY_ID		
Entity Name	STUDY_LOAD		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	TYPE_STUDY_LOAD
Data Type	Number(10,0)		
Not Null	YES		
Unique	NO		

Comments

Тип нагрузки

Attribute 'ID'

Caption	ID
Entity Name	STUDY_LOAD
Primary Key	YES
Foreign Key	NO
Data Type	Char(20)
Not Null	YES
Unique	NO

Comments

Первичный ключ

Entity 'Subjects'

Caption	Subjects

Primary Key 'pk_subjects'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number	YES	NO	NO	NO	Первичный ключ
	NAME		Varchar2(120)	YES	NO	NO	NO	Название дисциплины
FK	FACULTY_ID		Number	NO	NO	NO	NO	Факультет на который распределена нагрузка

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_FACULTY_SUBJECTS	Non-identifying	Faculty	Subjects	1:N
FK_SUBJECT_STUDY	Non-identifying	Subjects	STUDY_LOAD	1:N

Comments

Справочник дисциплин

Attribute 'ID'

Caption	ID
Entity Name	Subjects
Primary Key	YES
Foreign Key	NO
Data Type	Number
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'NAME'

Caption	NAME
Entity Name	Subjects
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(120)
Not Null	YES
Unique	NO

Comments

Название дисциплины

Attribute 'FACULTY_ID'

Caption	FACULTY_ID		
Entity Name	Subjects		
Primary Key	NO		
Foreign Key	YES	**Parent Entity**	Faculty
Data Type	Number		
Not Null	NO		
Unique	NO		

Comments

Факультет на который распределена нагрузка

Entity 'TYPE_STUDY_LOAD'

Caption	TYPE_STUDY_LOAD

Primary Key 'PK_TYPE_stady_load'

Attribute Name	type	Parent Entity
ID	PK	-----

Attributes

Key	Attribute Name	Domain	Data Type	N[1)]	U[2)]	C[3)]	D[4)]	Comments
PK	ID		Number(10,0)	YES	NO	NO	NO	Первичный ключ
	NAME		Varchar2(120)	YES	YES	NO	NO	Тип учебной нагрузки

Relationships

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_TYPE_STUDY	Non-identifying	TYPE_STUDY_LOAD	STUDY_LOAD	1:N

Keys

Key Type	Constraint Name	Attributes
PK	PK_TYPE_stady_load	ID
AK	NAME	NAME

Comments

Виды учебной нагрузки

Attribute 'ID'

Caption	ID
Entity Name	TYPE_STUDY_LOAD
Primary Key	YES
Foreign Key	NO
Data Type	Number(10,0)
Not Null	YES
Unique	NO

Comments

Первичный ключ

Attribute 'NAME'

Caption	NAME
Entity Name	TYPE_STUDY_LOAD
Primary Key	NO
Foreign Key	NO
Data Type	Varchar2(120)
Not Null	YES
Unique	YES

Comments

Тип учебной нагрузки

[1)]Not Null
[2)]Unique
[3)]Check
[4)]Default

Attribute Report

Attribute Name	Entity Name	Domain	Data Type	PK	FK
ID	Faculty		Number	YES	NO
NAME	Faculty		Varchar2(30)	NO	NO
FACULTY_ID	GROUPS		Number	NO	YES
HEAD_GROUP	GROUPS		Number	NO	YES
ID	GROUPS		Number	YES	YES
NAME	GROUPS		Varchar2(10)	NO	NO
GROUP_ID	LESSONS		Number	YES	YES
PREPOD_ID	LESSONS		Number	YES	YES
STUDY_ID	LESSONS		Char(20)	YES	YES
FACULTY_ID	Prepods		Number	NO	YES
FIRST_NAME	Prepods		Varchar2(120)	NO	NO
ID	Prepods		Number	YES	NO
LAST_NAME	Prepods		Varchar2(120)	NO	NO
DATE	RATING		Date	NO	NO
ID	RATING		Number	YES	NO
IS_ABSENT	RATING		Char(1)	NO	NO
PREPODS_ID	RATING		Number	NO	YES
STUDENT_ID	RATING		Number	NO	YES
STUDY_ID	RATING		Char(20)	NO	YES
VAL	RATING		Number	NO	NO
FIRST_NAME	Students		Varchar2(30)	NO	NO
GROUP_ID	Students		Number	NO	YES
ID	Students		Number	YES	NO
LAST_NAME	Students		Varchar2(60)	NO	NO
hours	STUDY_LOAD		Number	NO	NO
ID	STUDY_LOAD		Char(20)	YES	NO
SUBJECT_ID	STUDY_LOAD		Number	NO	YES
Type_STUDY_ID	STUDY_LOAD		Number(10,0)	NO	YES
FACULTY_ID	Subjects		Number	NO	YES
ID	Subjects		Number	YES	NO
NAME	Subjects		Varchar2(120)	NO	NO
ID	TYPE_STUDY_LOAD		Number(10,0)	YES	NO
NAME	TYPE_STUDY_LOAD		Varchar2(120)	NO	NO

Key Report

Key Type	Key Name	Entity Name
AK	NAME	Faculty
PK	PK_FACULTY	Faculty
PK	pk_group	GROUPS
PK	Key3	LESSONS
PK	PK_PREPODS	Prepods
PK	PK_RATING	RATING
PK	pk_students	Students
PK	PK_STUDY_LOAD	STUDY_LOAD
PK	pk_subjects	Subjects
AK	NAME	TYPE_STUDY_LOAD
PK	PK_TYPE_stady_load	TYPE_STUDY_LOAD

Key 'NAME'

Caption	NAME		
Key Type	Alternate Key		
Entity Name	Faculty		
Attribute		**Order**	**Foreign Key**
NAME		ASC	NO

Key 'PK_FACULTY'

Caption	PK_FACULTY		
Key Type	Primary Key		
Entity Name	Faculty		
Attribute		**Order**	**Foreign Key**
ID		ASC	NO

Key 'pk_group'

Caption	pk_group		
Key Type	Primary Key		
Entity Name	GROUPS		
Attribute		**Order**	**Foreign Key**
ID		ASC	YES

Key 'Key3'

Caption	Key3		
Key Type	Primary Key		
Entity Name	LESSONS		
Attribute		**Order**	**Foreign Key**
GROUP_ID		ASC	YES
PREPOD_ID		ASC	YES
STUDY_ID		ASC	YES

Key 'PK_PREPODS'

Caption	PK_PREPODS	
Key Type	Primary Key	
Entity Name	Prepods	
Attribute	**Order**	**Foreign Key**
ID	ASC	NO

Key 'PK_RATING'

Caption	PK_RATING	
Key Type	Primary Key	
Entity Name	RATING	
Attribute	**Order**	**Foreign Key**
ID	ASC	NO

Key 'pk_students'

Caption	pk_students	
Key Type	Primary Key	
Entity Name	Students	
Attribute	**Order**	**Foreign Key**
ID	ASC	NO

Key 'PK_STUDY_LOAD'

Caption	PK_STUDY_LOAD	
Key Type	Primary Key	
Entity Name	STUDY_LOAD	
Attribute	**Order**	**Foreign Key**
ID	ASC	NO

Key 'pk_subjects'

Caption	pk_subjects	
Key Type	Primary Key	
Entity Name	Subjects	
Attribute	**Order**	**Foreign Key**
ID	ASC	NO

Key 'NAME'

Caption	NAME		
Key Type	Alternate Key		
Entity Name	TYPE_STUDY_LOAD		
Attribute		**Order**	**Foreign Key**
NAME		ASC	NO

Key 'PK_TYPE_stady_load'

Caption	PK_TYPE_stady_load		
Key Type	Primary Key		
Entity Name	TYPE_STUDY_LOAD		
Attribute		**Order**	**Foreign Key**
ID		ASC	NO

Relationship Report

Relationship Name	Relationship Type	Parent Entity	Child Entity	Card.
FK_FACULTY_GROUP	Non-identifying	Faculty	GROUPS	1:N
FK_FACULTY_SUBJECTS	Non-identifying	Faculty	Subjects	1:N
FK_GROUP_LESSONS	Identifying	GROUPS	LESSONS	1:N
fk_group_of students	Non-identifying	GROUPS	Students	1:N
fk_head_of_group	Non-identifying	Students	GROUPS	1:N
FK_PREPOD_RATING	Non-identifying	Prepods	RATING	1:N
FK_PREPODS_LESSONS	Identifying	Prepods	LESSONS	1:N
FK_STUDENT_RATING	Non-identifying	Students	RATING	1:N
FK_STUDY_LESSONS	Non-identifying	STUDY_LOAD	RATING	1:N
FK_STUDY_PREPODS	Non-identifying	Faculty	Prepods	1:N
FK_SUBJECT_STUDY	Non-identifying	Subjects	STUDY_LOAD	1:N
FK_TYPE_STUDY	Non-identifying	TYPE_STUDY_LOAD	STUDY_LOAD	1:N
Relationship20	Identifying	STUDY_LOAD	LESSONS	1:N

Relationship Name 'FK_FACULTY_GROUP'

Caption	FK_FACULTY_GROUP		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Faculty		
Child Entity	GROUPS		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_FACULTY'	ID	PK_FACULTY

Relationship Name 'FK_FACULTY_SUBJECTS'

Caption	FK_FACULTY_SUBJECTS		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Faculty		
Child Entity	Subjects		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_FACULTY'	ID	PK_FACULTY

Relationship Name 'FK_GROUP_LESSONS'

Caption	FK_GROUP_LESSONS		
Relationship Type	Identifying	**Cardinality**	1:N
Mandatory Parent	YES		
Mandatory Child	NO		
Parent Entity	GROUPS		
Child Entity	LESSONS		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'pk_group'	ID	pk_group

Relationship Name 'fk_group_of students'

Caption	fk_group_of students		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	GROUPS		
Child Entity	Students		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'pk_group'	ID	pk_group

Relationship Name 'fk_head_of_group'

Caption	fk_head_of_group		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Students		
Child Entity	GROUPS		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'pk_students'	ID	pk_students

Relationship Name 'FK_PREPOD_RATING'

Caption	FK_PREPOD_RATING		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Prepods		
Child Entity	RATING		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_PREPODS'	ID	PK_PREPODS

Relationship Name 'FK_PREPODS_LESSONS'

Caption	FK_PREPODS_LESSONS		
Relationship Type	Identifying	**Cardinality**	1:N
Mandatory Parent	YES		
Mandatory Child	NO		
Parent Entity	Prepods		
Child Entity	LESSONS		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_PREPODS'	ID	PK_PREPODS

Relationship Name 'FK_STUDENT_RATING'

Caption	FK_STUDENT_RATING		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Students		
Child Entity	RATING		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'pk_students'	ID	pk_students

Relationship Name 'FK_STUDY_LESSONS'

Caption	FK_STUDY_LESSONS		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	STUDY_LOAD		
Child Entity	RATING		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_STUDY_LOAD'	ID	PK_STUDY_LOAD

Relationship Name 'FK_STUDY_PREPODS'

Caption	FK_STUDY_PREPODS		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Faculty		
Child Entity	Prepods		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_FACULTY'	ID	PK_FACULTY

Relationship Name 'FK_SUBJECT_STUDY'

Caption	FK_SUBJECT_STUDY		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	Subjects		
Child Entity	STUDY_LOAD		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'pk_subjects'	ID	pk_subjects

Relationship Name 'FK_TYPE_STUDY'

Caption	FK_TYPE_STUDY		
Relationship Type	Non-identifying	**Cardinality**	1:N
Mandatory Parent	NO		
Mandatory Child	NO		
Parent Entity	TYPE_STUDY_LOAD		
Child Entity	STUDY_LOAD		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_TYPE_stady_load'	ID	PK_TYPE_stady_load

Relationship Name 'Relationship20'

Caption	Relationship20		
Relationship Type	Identifying	**Cardinality**	1:N
Mandatory Parent	YES		
Mandatory Child	NO		
Parent Entity	STUDY_LOAD		
Child Entity	LESSONS		

Referential Integrity

	Delete
Parent	NO ACTION

Keys

Key Type	**Parent Key**	**Child Key**
Primary Key 'PK_STUDY_LOAD'	ID	PK_STUDY_LOAD